

EDITORIAL FOCUS
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Inherited burdens: the cardiac cost of parental obesity in offspring following myocardial infarction

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Obesity is a major health concern worldwide, especially in the United States, where rates continue to rise. It is estimated that over 1 billion people worldwide are obese, including more than 100 million individuals (>40% of adults) in the United States. Obesity is characterized by excessive body fat accumulation. Not only is obesity linked to metabolic disorders, but it also increases the risk of various chronic diseases, including cardiovascular diseases. Moreover, parental obesity increases their offspring's risks of obesity, diabetes mellitus, asthma, infection, stroke, neurocognitive disorders, and coronary heart disease (1). However, the underlying mechanisms remain incompletely understood.

In this journal, do Carmo et al. (2) examined the impact of parental obesity on offspring in the context of acute myocardial infarction (MI). In this study, Sprague-Dawley rat parents were fed a high-fat diet (HFD), and their offspring were fed a chow diet from weaning. Control rats were offspring from lean parents without exposure to the HFD. MI was induced by permanent ligation of the left anterior descending coronary artery. At baseline (no MI), the offspring of obese parents are lean but show signs of diastolic dysfunction, whereas the systolic function remains normal. Compared with controls, offspring from obese parents exhibit increased ventricular fibrillation, worse cardiac dysfunction, and reduced survival following MI, despite no difference in infarct size. Cardiac fibers from the offspring of obese parents display reduced oxygen consumption rate and mitochondrial respiration post-MI, but increased mitochondria-derived superoxide production, indicative of impaired mitochondrial metabolism and function. In addition, the male offspring of obese parents, at baseline, show lower left ventricular levels of connexin 43, a protein that forms gap junctions and is essential for propagation of electrical impulses. This may, at least partially, explain why offspring of obese parents show higher ventricular fibrillation following MI. Overall, this study provides the first evidence that parental obesity negatively affects their offspring's response to MI. It draws attention to the broader health risks of obesity, particularly its long-lasting effects across generations.

This study by do Carmo et al. (2) evaluated the impact of parental obesity on cardiac function, survival, and

mitochondrial function after MI in offspring; however, the effect on cardiac remodeling, including inflammation, angiogenesis, and scar formation, was not investigated. In terms of mechanisms, this study focused on cardiomyocyte changes as the potential mechanism, but alterations in other cells were not evaluated. In addition to the mechanisms revealed by the current study, we propose that the following changes induced by obesity, including epigenetic modifications, trained immunity, and alterations in gut microbiome, very likely contribute to worse MI outcomes in offspring of obese parents (Fig. 1).

Epigenetic modifications are defined by chemical changes to DNA and histone proteins that influence gene expression without altering the DNA sequence. The modifications include DNA methylation, histone modification (e.g., methylation, acetylation, phosphorylation, and ubiquitination), as well as the expression of noncoding RNAs, all of which can cause chromatin remodeling, modulate accessibility of DNA to transcriptional machinery, and subsequently affect gene expression. Epigenetic modifications are considered a key mechanism linking parental obesity with offspring outcomes. A recent study demonstrates that both human and mouse adipose tissues retain cellular transcriptional changes after weight loss (3). This memory largely depends on stable epigenetic changes in adipocytes and probably other cell types (e.g., endothelial cells and immune cells). After weight loss, adipocytes still maintain an upregulation of inflammatory and extracellular matrix remodeling-related pathways and a downregulation of adipocyte-specific metabolic pathways, such as fatty acid omega oxidation, fatty acid biosynthesis, and adipogenesis. Not surprisingly, mice carrying this obesogenic memory demonstrate accelerated rebound weight gain in response to HFD feeding. Epigenetic modifications can be passed down to future generations, a phenomenon known as transgenerational epigenetic inheritance. It is shown that maternal Western-type diet exacerbates atherosclerosis development in offspring by inducing activator protein 1 (AP1)-associated epigenetic memory in aortic endothelial cells, which increases chromatin accessibility to inflammatory genes and induces a heightened inflammatory response (4). Whether changes in cardiac fibers of offspring from obese parents revealed in this study are related to epigenetic



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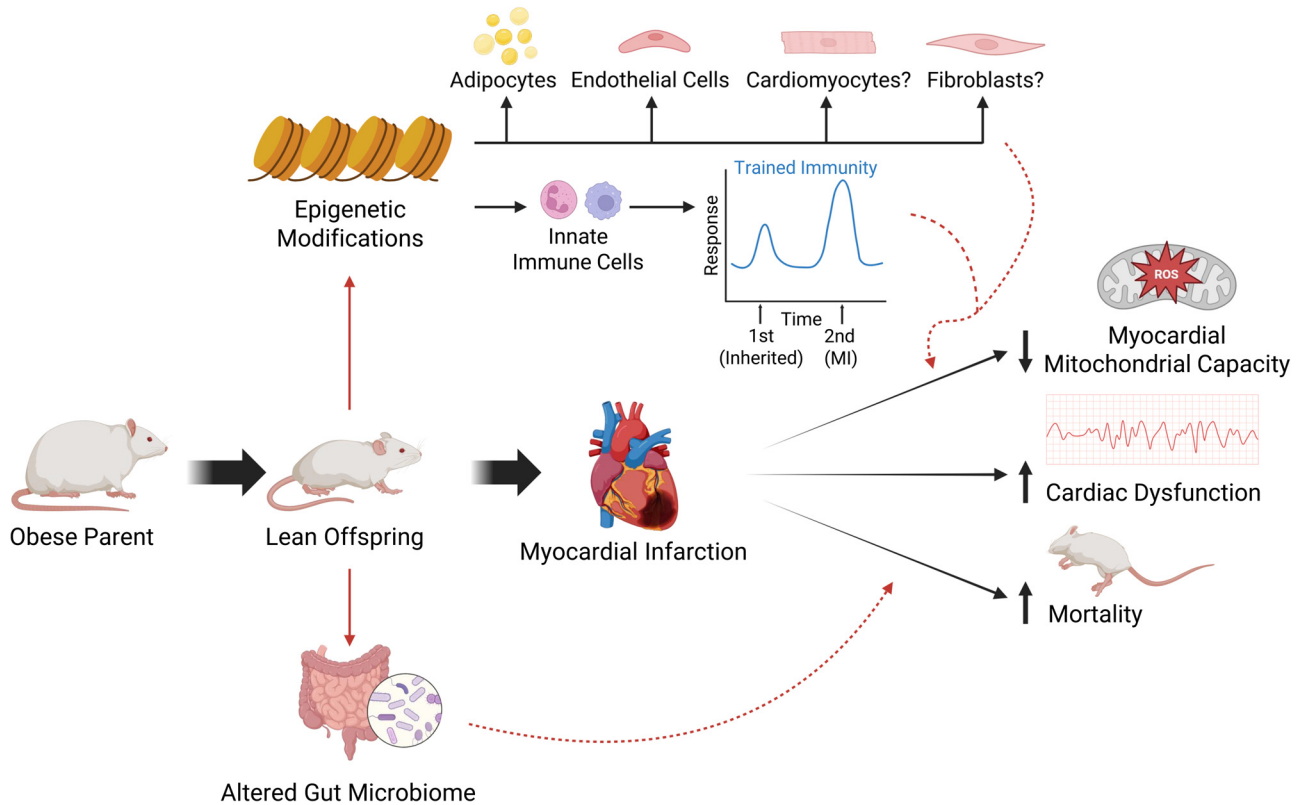


Figure 1. Diagram showing how parental obesity affects offspring's response to myocardial infarction (MI) injury. Offspring from obese parents may exhibit epigenetic modifications in adipocytes, endothelial cells, and probably cardiomyocytes and fibroblasts, trained immunity in macrophages and neutrophils, as well as reduced diversity of gut microbiome. When offspring from obese parents are subject to MI, these memories may confer a heightened inflammatory response that exacerbates adverse cardiac remodeling, worsens cardiac function, and increases mortality. The dashed line indicates where more evidence is needed. ROS, reactive oxygen species. Figure created with a licensed version of BioRender.com.

modifications remains unknown. Therefore, future studies are warranted to examine whether the epigenetic memories in cardiomyocytes, endothelial cells, and fibroblasts transmitted to offspring of obese parents exacerbate cardiac injury post-MI.

Trained immunity, a type of innate immune memory, is characterized by a stronger innate immune and inflammatory response upon exposure to a second same or different insult compared with the first stimulus. Epigenetic modifications contribute to the development and maintenance of trained immunity. Accumulating evidence has shown that obesity can induce trained immunity, which exacerbates inflammation and disease progression in different conditions. For instance, obesity triggers persistent reprogramming of the innate immune system, which lasts long even after normalization of body weight. Compared with controls, mice with a past history of obesity exhibit elevation of inflammation, aggravation of pathological retinal angiogenesis, and neuronal degeneration in mouse models of laser-induced retinal injury (5). Adipose tissue transplantation, bone marrow transfer, and utilization of mice with specific macrophage depletion confirm that these effects are mediated by both adipose tissue macrophages (ATMs) and resident retinal microglia. After injury, ATMs can infiltrate choroidal neovascularization lesions and contribute to disease progression. In addition, ATMs from former obese mice exhibit persistent epigenomic reprogramming toward enhanced angiogenic and inflammatory responses.

Another study reveals that an alternating HFD surprisingly accelerates atherosclerosis, compared with a continuous HFD (6). Mechanistically, alternating HFD induces trained immunity, and previously HFD-exposed mice produce more neutrophils after re-exposure to the HFD. Neutrophils infiltrate plaques and exacerbate atherosclerosis by releasing neutrophil extracellular traps and producing pro-inflammatory cytokines. Trained immunity has been shown to be transmitted to the subsequent two generations, and the progeny of trained mice that survived a fungal infection shows improved protection against bacterial infections (7). The bone marrow progenitors and monocytes from offspring of trained mice display transcriptional and epigenetic alterations associated with an upregulation of and more access to pro-inflammatory genes. Given the indispensable roles neutrophils and macrophages play in postischemic inflammation and cardiac remodeling (8), it is reasonable to speculate that obesity-induced trained immunity may be passed on to the offspring, which induces a heightened innate immune and inflammatory response following MI and impedes MI recovery.

The gut microbiota plays a crucial role in maintaining immune homeostasis and repair capacity. Loss of gut microbiota has been shown to increase cardiac rupture and mortality after MI (9). The underlying mechanisms are related to the inhibition of short-chain fatty acid (SCFA) production and monocyte infiltration into the ischemic site, as dietary SCFA supplementation or monocyte transplantation partially

rescues the worse MI outcomes induced by microbiota depletion. MI per se is also associated with broad alterations in the gut microbiota, including loss of a beneficial bacterial population, *Lactobacillus*; supplementation with a *Lactobacillus* probiotic in microbiota-depleted mice improves cardiac function after MI. Furthermore, mice receiving fecal transplants from MI donors have worse post-MI cardiac dysfunction than those receiving fecal transplants from non-MI mice. These findings indicate that the gut microbiome is essential for cardiac repair post-MI. Parental high-fat intake reduces the diversity of the gut microbiome in offspring, which elevates disease susceptibility. Offspring of obese parents exhibit greater mortality or more severe manifestations in *Escherichia coli* infection, experimental autoimmune encephalitis, and oral peanut sensitization models (10). Cohousing offspring of obese mice with normal mice rescues their susceptibility to *Escherichia coli* infection and immune dysregulation, implying that altered gut microbiota contributes to immune dysfunction. Therefore, it would be interesting to know whether altered gut microbiota in offspring of obese parents contributes to worse MI outcomes.

In summary, this study by do Carmo et al. (2) offers valuable insights into the far-reaching effects of obesity. By using a multifaceted approach, the study demonstrates that parental obesity adversely influences offspring's response to cardiac ischemic injury, featured by worse cardiac function and increased mortality. These findings provide a strong foundation for future research on intergenerational health effects and suggest potential targets for therapeutic intervention.

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DISCLAIMERS

The content is solely the authors' responsibility and does not necessarily represent the official views of the University of South Florida.

DISCLOSURES

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AUTHOR CONTRIBUTIONS

M.D.C. and Y.M. prepared figures; drafted manuscript; edited and revised manuscript; and approved final version of manuscript.

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